SALVIN
Engineering - Processing - Packaging

ONION - GARLIC DEHYDRATION PLANT

Your Trusted Partner for Consultancy and Turnkey Project Execution

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COMPANY OVERVIEW

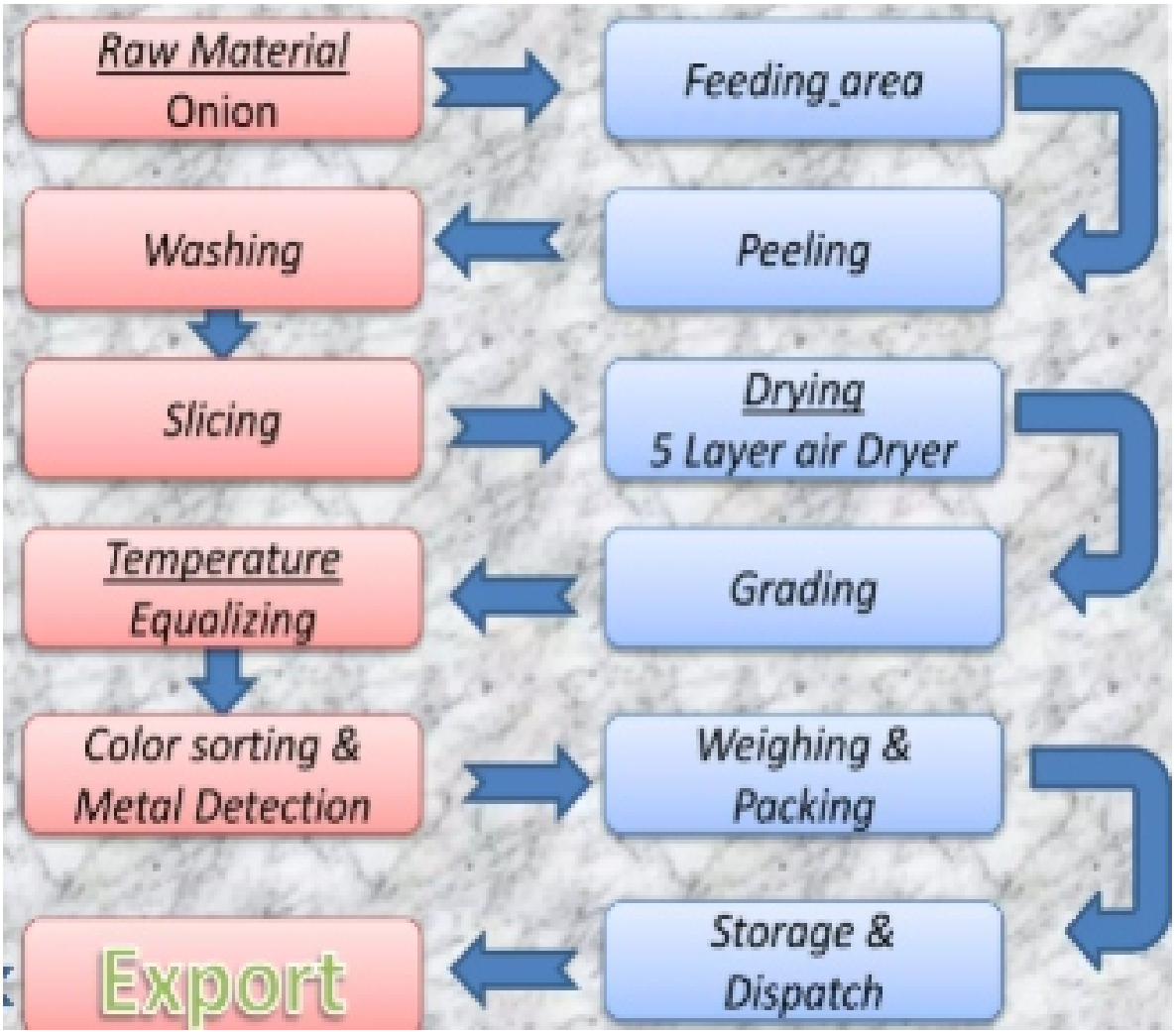
Salvin Industries is a leading engineering, technology and consulting conglomerate, working closely with diversified industries spanning across food processing, agriculture, plastics, chemicals, metals healthcare, packaging, Dealer, distributer, super stocks, marketing plan, product analysis and other industries and business sectors offering complete turn-key solutions including advanced engineering, Industry 4.0 solutions, digital transformation and expert technical and management consultancy. With 10+ years of experience and global presence in 8+ countries, Salvin Industries is serving from the most demanding to the most distant customers across the globe. We empower them for rewriting their future with our depths of knowledge, unique ideas, technical expertise, creative solutions and enduring results, all at the most optimal costs, right at their doorsteps.

Armed with the eminence in technological expertise, global capabilities and a process driven approach, we have the ability to take on any challenging project. Crossing the geographical borders & cultures and living up to our core values Excellence, Respect, Care & Joy, we propel innovation to engineer solutions that build & foster equality. Working together as one team with all our business partners & associates, we are transforming industries, fueling economies, developing sustainable solutions and thereby elevating societies, globally. We realize the need to respond to the dynamism & speed with which the world is changing all around us and understand the need to anticipate

ONION - GARLIC DEHYDRATION PLANT

In a global market increasingly driven by convenience and long-term food security, an Onion and Garlic Dehydration Plant serves as a vital bridge between seasonal agriculture and the year-round industrial food chain. These facilities utilize sophisticated thermal processing technologies, such as continuous belt dryers and automated slicing systems, to extract moisture while locking in the volatile oils responsible for the crops' signature pungency and nutritional profile. By converting bulky, perishable bulbs into lightweight, concentrated flakes or powders, the plant effectively eliminates the risk of post-harvest spoilage and significantly reduces logistics overhead. Modern plants, often constructed from food-grade stainless steel and managed via PLC-controlled automation, ensure that the final product meets rigorous international hygiene standards, making it an essential asset for supplying the spice, snack, and pharmaceutical sectors.

PROCESS DIAGRAM OF ONION - GARLIC DEHYDRATION PLANT



ONION - GARLIC DEHYDRATION PLANT

THE PROCESS

Washing

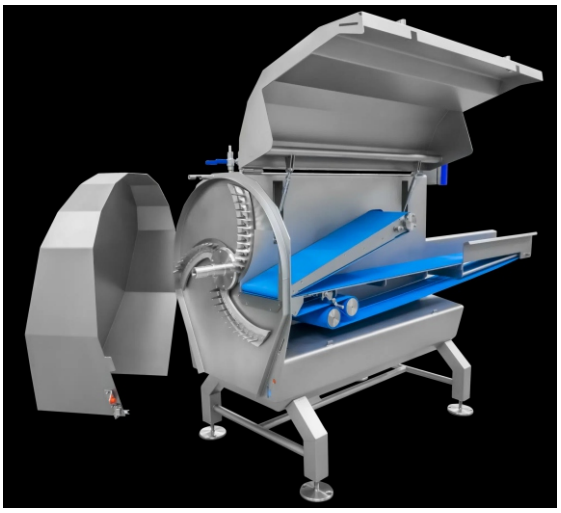
Raw onions and garlic bulbs are fed into a stainless steel washing tank equipped with a high-pressure air bubbling system. Compressed air forced through manifolds creates turbulence, causing bulbs to strike each other while clay, dust, and surface contaminants are loosened and removed. Food-grade PVC or SS wire mesh conveyor carries material continuously. High-pressure blower rated 2.2–3.7 kW generates agitation. Water consumption 300–1000 litres per hour depending on capacity. Capacity range 200–2000 kg per hour. Drain system permits frequent water changes

Topping & Tail Cutting Machine

For garlic, dry peeling is accomplished using compressed air and rubber rollers that rupture and remove papery skin without water, achieving peeling rate above 95%. No pollution to cloves. Fully automatic operation with adjustable peeling layers. For onions, topping and tailing units remove root and stem ends before slicing. Constructed of food-grade SS304 with automatic feeding system. Simple hygienic design, easy to clean and disinfect

Slicer Machine /Cutter Machine

Cleaned bulbs are conveyed to high-capacity slicing machines fitted with stainless steel cutting blades. Machine performs shredding, mincing, slicing, or dicing in uniform thickness. Contact parts SS304 with MS/aluminum base structure. Capacities available: 2000 kg/hr (3HP), 3500 kg/hr (5HP), 5000 kg/hr (5HP). Compact dimensions from 1.2m x 1.2m x 1.5m to 2m x 1.5m x 1.5m. Portable, noise-free working, sturdy design, hassle-free operation



ONION - GARLIC DEHYDRATION PLANT

THE PROCESS

5-Stage Conveyor Dryer

Sliced onions or garlic are evenly distributed onto stainless steel mesh belts (12–60mm mesh opening) arranged in vertical planes across 5 layers. Each belt has independent drive and variable speed control. Hot air generated by steam heating coils or heat exchanger-based hot air generator rises through permeable bed of material. Drying temperature range: 80–120°C for initial stages, reducing to 60–80°C for final stages. Residence time adjustable; final moisture reduced to approximately 8%. Exhauster fans on top release moist air. Heating coils enable precise temperature control. Thermal efficiency 60–75%. Suitable for continuous 24/7 operation.

Elevator / Screw Conveyor

Dried onion flakes or garlic granules are transferred gently to the grinding section via enclosed bucket elevators or screw conveyors. Smooth transport prevents breakage and dust escape. Ss304 construction, hygienic design .

Grinding Mill / Pulverizer

Dried material is fed into a high-speed grinding chamber. Rotating hammers or blades reduce particles to specified mesh sizes. Interchangeable screens enable production of 8–16 mesh, 16–26 mesh, 26–40 mesh, 40–80 mesh granules, or 80–100 mesh, 100–120 mesh fine powder. Power consumption 32–48 kW depending on model. Grinding roller contact length 1000–2000mm. Fully automatic operation

ONION - GARLIC
DEHYDRATION PLANT

THE PROCESS

Gyratory Sifter / Vibratory Sifter

Ground powder passes through multi-deck sifting screens to separate and classify particles by size. Oversize material is automatically returned to mill inlet for regrinding. Sifting area 10–17 square meters. Ensures uniform particle size distribution and product consistency. Dust-tight construction prevents airborne fines

Automatic Filling & Packaging Machine

Finished onion powder or garlic powder is conveyed to vertical form-fill-seal machine or auger filler. Precise dosing into pouches, sachets, or rigid containers. Optional nitrogen flushing for extended shelf life. Dust collector integrated to maintain clean environment. Date coder and checkweigher with automatic rejection available. Stainless steel construction, CE-certified options

LET'S WORK
TOGETHER

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